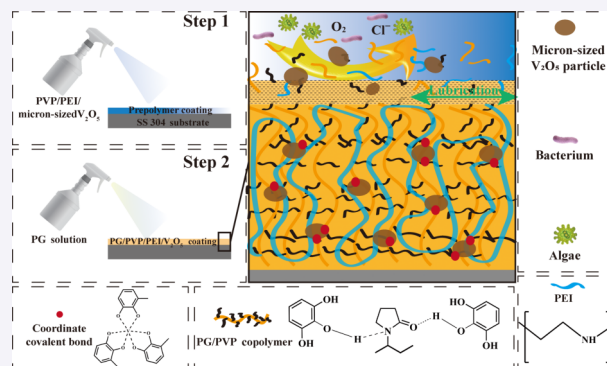


MAP-inspired dual crosslinked PVP-phenol sprayable hydrogel coating for stable marine antifouling applications

Haobo Shu¹, Junnan Cui¹, Yuhan Liu², Xiuqin Bai², Chengqing Yuan^{2,✉}, Pan Cao^{1,✉}

✉ Cite this article: Shu H, Cui J, Liu Y, et al. *Friction* 2025, **13**(11): 9440946. <https://doi.org/10.26599/FRICT.2025.9440946>

ABSTRACT: The application of antifouling paints to the surfaces of marine installations is the most economically efficient means for mitigating damage caused by marine biofouling in the shipping industry. However, conventional antifouling paints currently in widespread use can no longer meet the requirements of green antifouling. Although hydrogel coatings have made great progress in marine antifouling applications, current hydrogel coatings still suffer from construction difficulties and poor mechanical stability under wet conditions. In this paper, we innovatively exploit the phenomenon of the absorption of pyrogallol (PG) by large-molecular-weight polyvinylpyrrolidone (PVP), resulting in hydrophilic copolymer macromolecules, to propose a prepolymer-reactor rapid contact molding of sprayable hydrogel coatings. The PG/PVP copolymer produced microscopic reticular mimetic mussel adhesion protein (MAP) bioscaffolds via the chemical crosslinking of polyethyleneimine (PEI), contributed to the conversion of PG to PG-quinone upon the introduction of vanadium pentoxide particles, increased the hydrophobicity of the system and enhanced waterproof adhesion. The wet adhesion of the hydrogel coatings was measured up to 3.42 MPa via the micrometer scratch method, indicating that the prepared hydrogel coating had a stable adhesive force in a wet environment. The hydrogel coating was instantly molded on the surface of 304 stainless steel (SS) via two-step spraying. The swelling, friction, antifouling, and anticorrosion properties of the coatings were investigated along with the wet adhesion strength on the SS surfaces. The results showed that the hydrogel, after double cross-linking of PEI and V_2O_5 , had a swelling rate within 30% and a low modulus along with stable lubricating properties. After the formation of the hydrogel coating, the inhibition rate of common bacteria and algae in the ocean reached more than 99%, and the electrochemical corrosion protection rate of SS reached 63.49%. This study provided ideas for improving the wet adhesion of hydrophilic marine antifouling coatings.



KEYWORDS: hydrogel coating; polyvinyl pyrrolidone (PVP); antifouling; corrosion resistance; lubrication

1 Introduction

The process of the colonization of artificial marine substrates by microorganisms (bacteria, fungi, and diatoms) and macro-organisms (algae and invertebrates) is known as marine biofouling [1]. The occurrence of fouling can be divided into three parts [2]. Within hours or even minutes of immersion in seawater, biomolecules, such as proteins and polysaccharides, are adsorbed to the surface of the artificial marine substrate by interaction forces, such as van der Waals forces, hydrogen (H)-bonding, and electrostatic forces, forming a conditioning film. In the second stage, microorganisms colonize the conditioned film and secrete extracellular polymers consisting of polysaccharides, proteins, and glycoproteins, within a few days, providing a breeding ground for

further adsorption of microorganisms and the formation of mucus biofilms composed of water, biomolecules, and secretions. The third stage usually occurs after a few weeks, when the biofilm provides nutrients for the colonization of large fouling organisms and the slow formation of complex communities of organisms, leading to the formation of large-scale fouling after a few months [3].

Biofouling has plagued mankind for centuries as a major factor contributing to the failure of subsea installations [4]. From an economic perspective, biofouling reduces fuel efficiency by affecting surface roughness and overall weight [5, 6]. In addition, secretions from the reproduction and metabolism of fouling organisms, which are often acidic, accelerate surface corrosion of marine equipment, leading to increased maintenance costs [7–9]. From an environmental perspective, increased fuel consumption

¹ School of Mechanical Engineering, Yangzhou University, Yangzhou 225009, China. ² State Key Laboratory of Maritime Technology and Safety, Wuhan University of Technology, Wuhan 430063, China.

✉ Corresponding authors. E-mail: P. Cao, caopan@yzu.edu.cn; C. Yuan, ycq@whut.edu.cn

Received: January 17, 2024; Revised: May 31, 2024; Accepted: June 14, 2024

© The Author(s) 2025. This is an open access article under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0, <http://creativecommons.org/licenses/by/4.0/>).

means more greenhouse gas emissions and fouling organisms adhering to the ship's surface can affect biodiversity along the route. The use of antifouling paints on the surfaces of marine equipment is by far the most cost-effective method. Tributyltin (TBT) was widely used in antifouling paints for ships in the 1970s and was once considered the most successful antifouling paint because of its excellent antifouling properties. However, toxicological studies on TBT have shown that TBT achieves its antifouling effect by inhibiting the fertilization process and larval metamorphosis in marine organisms and that its application and leaching in large quantities can affect the population of nontarget organisms and even cause embryonic malformations in mammals [10, 11]. As a result, the International Maritime Organization (IMO), at its November 1999 meeting in London, adopted a global ban on the use of organotin paints on all ships from 2003 and a total ban on the use of TBT in marine antifouling paints from 2008 [12]. As a result, copper-based antifouling coatings have become the commercially preferred and most widely used antifouling coatings, accounting for up to 80% of the total usage. These coatings, which consist of a binder and copper-based antifouling agents, such as cuprous oxide and copper omadine, have a broad-spectrum bactericidal effect, but are also a major source of copper loading in the marine environment. The extensive use of copper-based antifouling agents often exposes marine organisms to high concentrations of copper [13]. As a broad-spectrum biocide, cupric oxide kills fouling organisms but also has toxic effects on other "nontarget" marine populations, and copper-containing particles from ships' antifouling paints have been shown to inactivate a number of enzymes in marine organisms [14]. Although the toxicity of copper-based antifouling paints is much lower than that of TBT, their use in antifouling paints is still under review in several countries, and it is uncertain whether they will be completely replaced in the future. Nontoxic, low surface energy antifouling paints based on silicone/fluoropolymer materials, such as silicone and fluorine-based coatings, have also been reported to be used in marine antifouling, which makes it difficult for fouling organisms to adhere to the surface or be easily washed off. However, owing to the over-reliance on seawater rinsing, these antifouling coatings still have significant drawbacks for ships that have to spend long periods in port and for fixed marine installations [15, 16]. Therefore, the development of environment-friendly antifouling coatings with stable antifouling properties under both stationary and moving conditions is of particular concern.

Hydrogel antifouling coatings are bionic surfaces that have emerged in recent years to take environmental concerns into account. For antifouling hydrogel coatings, adhesion stability in wet environments, which is reminiscent of that of mussel adhesion proteins (MAPs), needs to be a key consideration. The extremely strong adhesion of MAP is due mainly to the 3,4-dihydroxyphenylalanine (DOPA) phenol moiety enriched in its structure. This process is characterized by chelation with metals, the formation of irreversible metal complexes on the metal substrate surface, and the formation of strong H-bonding interactions with polar polymers. The water resistance of MAP, in contrast, is related to the substitution of tyrosine residues with DOPA and the subsequent quinone-based oxidation of DOPA [17, 18]. Recent studies have shown that mussels store iron and vanadium (V) ions in intracellular metal storage particles that form biomolecular complexes with catechol. Compared with current mainstream synthetic epoxy adhesives, biomimetic bioadhesives modeled on such proteins have the advantages of strong adhesion and excellent water resistance and do not cause damage to the marine environment. Since Waite and Tanzer [19]

first discovered that proteins isolated and extracted from mussel adhesive discs were rich in lysine and DOPA and had strong *in vitro* adhesive properties, MAP has been shown to have excellent applications in biomedicine, biomimetic materials, and other fields over the past decades.

Current applications of MAP-inspired adhesives can be summarized as the use of synthetic polymers to replace the protein backbone of MAP and branching catechin derivatives on the side chains to achieve MAP-like wet adhesion. Yang et al. [20] used selenium semicarbamate (SeCA) copolymerized with DOPA to produce an organic solvent-free coating on cardiovascular scaffolds made of 316L stainless steel (SS), which provides a favorable platform for the treatment of cardiovascular diseases. This study demonstrated that a strong adhesive made by replacing the MAP protein backbone components with polymers can be firmly bonded to the metal substrate surface, resulting in good water resistance and biocompatibility. In recent years, a large number of materials have been reported to mimic the MAP principle by grafting functional groups of phenolic moieties (e.g., DOPA and catechol) to form covalent coordination bonds with the substrate to achieve adhesion. North et al. [21] reported excellent results with poly(catechol)-styrene as an underwater adhesive. Subsequently, Wang et al. [22] have obtained an environment-friendly bio-strong adhesive with bond strength comparable to that of commercial epoxy adhesives by simply mixing pyrogallol gallic acid with polyethyleneimine and endogenous oxygen provided by nano-SiO₂ groups to facilitate curing. This study also revealed that the addition of specific components to the bio-adhesive resulted in an increase in the overall bond strength. In addition, Jones and Wilker [23] added phosphate groups to catechol or DOPA bioglues, and the resulting polycatechol phosphate polymers showed a positive correlation between the amount of phosphate added in terms of dry bond strength. Both experiments showed that the addition of appropriate groups to biomimetic glues can help improve the adhesive properties and extend their range of applications. In addition, Gan et al. [24] developed an adhesive hydrogel based on silver-lignin nanoparticle-triggered dynamic redox catechol chemistry and introduced a persistent redox system into the hydrogel system for long-term adhesion. This demonstrated that the addition of specific moieties enhanced the performance of biomimetic hydrogel materials. Inspired by the dynamic amine-catechol interactions of mussel foot proteins, Shui et al. [25] mimicked the multidirectional fiber bundles of connective tissues to produce a dynamic hydrogel material with extremely strong isotropic tensile strength, ultra-high toughness, and resilience. With good underwater adhesion and electrical conductivity, the MAP biomimetic hydrogel has excellent mechanical stability. However, few studies have investigated the use of biomimetic MAP hydrogels for marine antifouling coatings.

When MAP bionic hydrogels are applied in practice for marine antifouling, construction convenience, molding difficulty, and cost are unavoidable issues. In this study, a two-step spray-molding hydrogel was designed by combining the principles of the adsorption of phenol by polyvinylpyrrolidone (PVP) and Michael addition and Schiff base reactions between polyethyleneimine (PEI) and phenol, as shown in Fig. 1. Pyrogallol (PG), one of the phenols with strong antimicrobial effects, was selected as the phenol source for hydrogel construction and, at the same time, saved cost by eliminating the need for other antifouling additives [26, 27]. In addition, the mechanical stability of the hydrogel was further improved by introducing V⁵⁺ into the hydrogel system to provide coordination crosslinking, thus mimicking the chelation principle of MAP with metal ions. The reaction mechanism

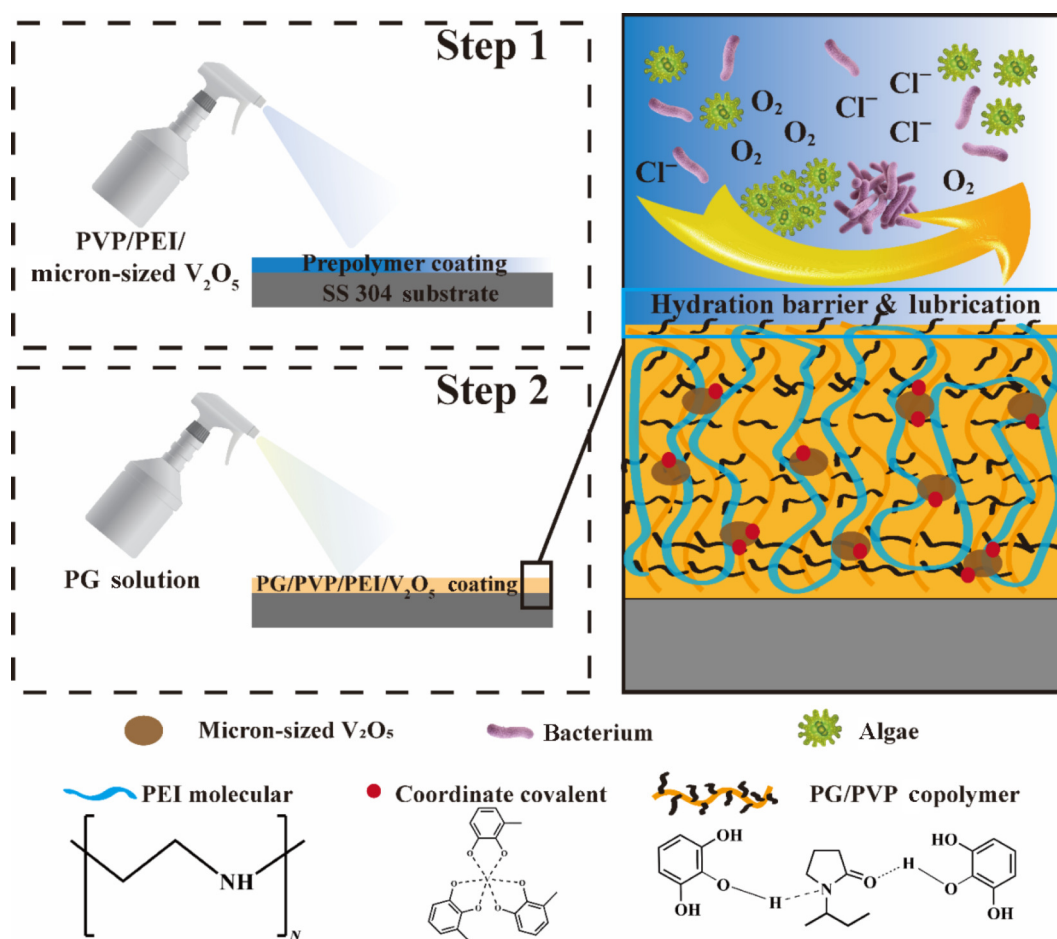


Fig. 1 Schematic diagram of preparation process of hydrogel coating.

between PG and PVP was investigated while characterizing the hydrogel, and a series of experiments were designed to characterize the mechanical stability, antifouling, anticorrosion, and lubricating properties of the hydrogel coatings. These results indicated that spray-formed PG/PVP/PEI/V₂O₅ coatings provided a solid evidence for the development of stable and easy-to-apply hydrogel coatings.

2 Experimental

2.1 Materials

Propidium iodide (PI) and glutaraldehyde were purchased from Aladdin Biochemical Technology Co., Ltd., China). PG, micron-sized vanadium pentoxide (VP), and PEI (99%, 1,800 MW) were purchased from Shanghai Macklin Biochemical Technology Co., Ltd., China. PVP (K90), crystal violet, and absolute ethanol (> 99.7%) were purchased from Sinopharm Chemical Reagent Co., Ltd., China. Luria-Bertani (LB) broth culture medium and LB agar culture medium were purchased from Shanghai Bio-way Technology Co., Ltd., China. 2216E liquid culture medium, 2216E agar culture medium, and BG11 culture medium were purchased from Qingdao Hope Biotechnology Co., Ltd., China. Sodium polycarboxylate dispersant (Ecospert 5040) was purchased from Shandong Usolf Chemical Technology Co., Ltd., China. The above reagents were of analytical grade and were used without further purification.

2.2 Synthesis of hydrogel prepolymer

PVP/PEI prepolymers were prepared by directly mixing PVP with

PEI solutions. First, PVP was prepared as a solution at a concentration of 50 mg/mL by heating at 80 °C and 800 r/min, and the solution was then placed in a vacuum drying oven to completely eliminate air bubbles after sufficient dissolution. PEI at a concentration of 99 vol% was diluted to a 50 vol% solution, mixed with the previous PG solution at a volume ratio of 1/10 and then loaded into a spray to obtain the PVP/PEI prepolymer. To obtain the PVP/PEI/V₂O₅ prepolymer, 0.1 vol% sodium polycarboxylate dispersant was added to the PVP/PEI prepolymer for modification. Then, 1 wt% micron-sized VP powder was added, and the mixture was ultrasonically vibrated to obtain a homogenous dispersion.

2.3 Preparation of hydrogel block and hydrogel film

In the experimental design, hydrogels were fabricated into films and blocks as required by characterization conditions. In addition to the prepolymers mentioned above, PG was prepared as a solution at 1 M concentration. For the preparation of PG/PVP and PG/PVP/PEI/V₂O₅ gels, the corresponding prepolymers were mixed directly with the PG solution, formed, rinsed with deionized water to remove unreacted materials, and then cut into the desired shapes. To obtain a hydrogel film-modified SS surface, the prepolymer and PG solutions were sprayed step by step via the two-step spraying method, and flocculent gel formation was observed, followed by rinsing with deionized water (Fig. 1). After drying with a blower, the above spraying process was repeated to obtain the desired thickness.

2.4 Basic characterization

The surface's three-dimensional (3D) morphology, both before

and after the hydrogel film swelled, was observed using a Keyence 3D laser scanning microscope (VK-X1000, Keyence Corp., Japan). In addition, the contact angle of pure water on the sample surface was measured via a contact angle measuring instrument (JC2000D, Shanghai Zhongchen Digital Technology Equipment Co., Ltd., China), and the average value of three different positions on the sample surface was obtained. An S-4800 field emission scanning electron microscope (SEM; Hitachi Instruments Inc., Japan) was used to collect the surface morphology of the coated samples.

The infrared spectra of the coated surfaces were acquired via a Cary 610/670 Fourier transform infrared spectrometer (FT-IR; Agilent Technologies Inc., USA) equipped with diamond attenuated total reflection (ATR) accessories. A total of 128 scans were used in the range of 4,000–400 cm^{-1} and a resolution of 4 cm^{-1} , with an average of 64 scans. The baseline was corrected, and CO_2 peaks were eliminated via Resolution Pro software. X-ray photoelectron spectroscopy (XPS) analysis was conducted with an ESCALAB 250Xi instrument (Thermo Fisher Scientific Inc., USA). The surface X-ray diffraction (XRD) data of the PG-PVP/PEI/ V_2O_5 samples were additionally examined via a Bruker D8 instrument (Bruker Corp., USA) equipped with $\text{Cu/K}\alpha$ radiation.

2.5 Hydrogel swelling test

The water content of a hydrogel can be quantified via a swelling test. As hydrogels form a dissolved layer on the surface after they absorb water and swell, the swelling rate (SR%) affects the physical and chemical properties of the hydrogel, the antifouling performance and the self-polishing rate to a certain extent. In addition, when used as an antifouling coating for ships, the weight of the coating is closely related to the total energy consumption, such that the swelling rate is one of the important indicators of hydrogel coatings. The experimental procedure was as follows.

The hydrogel coating was prepared according to the above method, rinsed with deionized water, placed in a refrigerator below -80°C for 1 h, placed in a vacuum freeze dryer for 48 h, and then placed in a vacuum drying oven at 40°C for storage until the hydrogel coating was dried to a constant weight, which was recorded as M_0 . The freeze-dried hydrogel blocks were then immersed in 250 mL of artificial seawater and allowed to swell at room temperature. During the first day of the experiment, samples were taken out every 2 h and weighed after the surface water was dried with filter paper. The weighing period was extended to 12 h, and the weight was recorded as M_t . The SR% was calculated via Eq. (1), and when the hydrogel coatings reached swelling equilibrium, the decreasing trend in the SR% was used to represent the self-polishing rate of the coating.

$$\text{SR}\% = \frac{M_t - M_0}{M_0} \times 100\% \quad (1)$$

2.6 Characterization of lubricating properties

The lubricating properties of the PG/PVP, PG/PVP/PEI, and PG/PVP/PEI/ V_2O_5 hydrogels were evaluated via a TriboStudio friction and wear tester (Yangzhou University, China) assembled with 6 mm diameter GCr15 balls as upper specimens. The dehydrated hydrogel was cut into squares of approximately $1\text{ mm} \times 1\text{ mm} \times 0.5\text{ mm}$, encapsulated in epoxy resin and placed in a mold to make a friction specimen, and the surface layer was fully sanded with sandpaper after the epoxy resin cured until the absorbed epoxy resin was polished clean. The friction samples were then placed in artificial seawater to dissolve completely, and the excess was removed with a knife. To simulate the

computational fluid dynamics (CFD) model of the material in real-life scenarios, 20 MPa was chosen as the average contact pressure to simplify the design, and the frictional load at this point was calculated to be 2 N via Hertzian contact theory [28, 29]. The specimens were clamped, and then friction was applied via reciprocal friction at frequencies of 1, 2, and 4 Hz with a reciprocal stroke of 5 mm using a constant normal pressure of 2 N. The coefficients of friction (COFs) were recorded, and friction was calculated via Hertzian contact theory at frequencies of 1, 2, and 4 Hz. Each set of tests was repeated at least three times at room temperature.

2.7 Mechanical performance test

2.7.1 Tensile test

Tensile testing of the hydrogel materials making up the coatings was carried out via an INSTRON 68TM-30 universal testing machine (Instron Corp., USA) equipped with a high-precision transducer. The hydrogel was cut into standard dumbbell-shaped tensile specimens with a thickness of 1 mm and a section of 4 mm after absorbing water to reach swelling equilibrium. When the samples were clamped, protective paper was wrapped around the two segments of each specimen to prevent the fixture from breaking the hydrogel. In the experiments, the loading accuracy was set to 0.5%, and the tensile speed was 30 mm/min.

2.7.2 Micro-scratch test

The mechanical stability of the hydrogel coatings was tested via a microscratch test (MST) with an Anton Paar CSM RST large load scratch tester (Anton Paar GbmH, Austria) with a spherical indenter ($r = 500\text{ }\mu\text{m}$); the test parameters are shown in Table 1.

According to Hertz's theory, the contact area of the spherical indenter with the stylus follows a semielliptical shape, at which point the contact pressure p_m is calculated via Eq. (2) [30]:

$$p_m = \frac{2}{3} \left(\frac{6L_c E_r^2}{\pi^3 R^2} \right)^{\frac{1}{3}} \quad (2)$$

where L_c is the critical normal force when the coating is destroyed; R is the radius of the spherical indenter; and E_r is the discounted modulus of elasticity, which is used to characterize the effect of the non-rigid indenter on the load-displacement and is introduced by Poisson's ratio and calculated via Eq. (3) [31]:

$$\frac{1}{E_r} = \frac{1 - \nu^2}{E} + \frac{1 - \nu_i^2}{E_i} \quad (3)$$

where E and ν are the Young's modulus and Poisson's ratio for the specimen respectively, and E_i and ν_i are the same parameters for the indenter. In this scratch test, the indenter material is diamond, and the sample substrate is 304 stainless steel. The values of Young's modulus and Poisson's ratio are taken from Table 2.

Table 1 Micro-scratch test parameters

Parameter	Value
Acquisition rate (Hz)	30
Begin load (mN)	1,000
End load (mN)	5,000
Loading rate (mN/min)	19,600
Speed (mm/min)	2
Length (mm)	5
Acoustic emission sensitivity	9

2.8 Marine anti-fouling performance test

The bacterial inhibitory properties of the coatings were evaluated using tests for typical Gram-positive bacterium, *Staphylococcus aureus* and a Gram-negative bacterium commonly found in the sea, *Vibrio natriegens*. *S. aureus* was cultured in LB broth medium, and *V. natriegens* was cultured in 2216E liquid medium. For bacterial adhesion tests, the samples were separately immersed in the appropriate liquid medium inoculated with the above bacteria at 37 °C for 24 h, washed with sterile phosphate-buffered saline (PBS) buffer, and placed in 2.5% glutaraldehyde solution at 4 °C for 4 h. The apoptotic bacteria fixed on the surface of the samples were then stained with a sterile PBS solution containing PI at a concentration of 50 µg/mL in a light-free environment, and the bacteria adhering to the surface after staining were observed at excitation and emission wavelengths of 535 and 615 nm, respectively. The treated coatings were placed on slides and used to observe images of bacterial adhesion on the prepared surfaces under a TCS SP8 STED confocal laser microscope (CLSM; Leica Microsystems GmbH, Germany).

The resistance of the coated surfaces to algal adhesion was tested, employing two common types of algae, namely, *Chlorella pyrenoidosa* and *Phaeodactylum tricornutum*. These algae were cultured in BG11 medium, while *P. tricornutum* was specifically cultured in silicon-enriched Guillard's F/2 medium. To perform the algal adhesion test, samples were immersed in the corresponding liquid medium with the above algae in a illuminated incubator and then removed after 7 d of immersion at 25 °C under a 12 h/12 h diurnal action environment. Poorly adhering algae on the surface were rinsed with sterile artificial seawater for 3 min. Collected rinsed samples were immersed in 2.5% glutaraldehyde fixative at 4 °C for 4 h under refrigeration to fix the surface algae and then placed on slides for observation. The chlorophyll contained in the algae could be observed at excitation wavelengths of 400–500 nm, and CLSM was used to observe algal adhesion to the sample surfaces.

The adhesion rate was quantitatively analyzed via ImageJ software and calculated via Eq. (4):

$$\text{AR}\% = \frac{A'}{A} \times 100\% \quad (4)$$

where AR% denotes the attachment rate; A' represents the area of microbial attachment; and A represents the total area of the field of view.

A plate count method was also used to test the inhibitory effect of the hydrogel coating on the adhesion of the above two bacteria. The samples before and after coating were placed in 24-well plates (one sample per well), and the surfaces of the samples were uniformly coated with liquid media containing the above bacteria. After incubation at 37 °C for 5 h, the samples were washed three times with 0.9% NaCl solution, and then, the bacteria were precipitated by centrifugation at 4,000 r/min for 15 min. The resuspended bacterial mixture (in 1 mL of 0.9% NaCl) was plated on LB agar and 2216E agar plates and incubated for 24 h. The antifouling properties of the samples were evaluated by observing bacterial growth on agar plates.

2.9 Electrochemical property analysis

Preparation of hydrogel-modified working electrodes included the use of a 0.5 mm² square copper wire welded to the surface of a 10 mm × 10 mm × 1 mm SS sheet, with ultrasonic cleaning of the residual resin and flux with anhydrous ethanol. After cleaning, the SS sheet welded with copper wires was encapsulated in epoxy resin, and a 1.0 cm² working surface was polished after the resin cured. The working surface was coated with a hydrogel coating using the above-mentioned spraying method, and the entire hydrogel-modified electrode was obtained after freeze-drying. Notably, the hydrogel-modified electrode must be completely dissolved in artificial seawater to simulate the actual environment of use as a working electrode. Foreelectrochemical experiments, a 10 mm × 10 mm × 0.1 mm platinum sheet electrode was used as the auxiliary electrode, and a saturated calomel electrode as the reference electrode to form a traditional three-electrode system. The corrosion resistance of the samples was evaluated via electrochemical experiments using a Gamry Interface 1010E electrochemical workstation (Gamry Instruments Inc., USA) immersed in a 3.5% NaCl solution. Electrochemical impedance spectroscopy (EIS) tests were carried out in the frequency range of 10 mHz–100 kHz with sinusoidal voltage variations of 10 mV. ZView2 software was used to analyze the EIS results of the equivalent circuit.

Potential polarization curves were obtained by scanning at a rate of 5 mV/s in the range of −0.5 to 0.5 V at room temperature. The corrosion potential (E_{corr}) and corrosion current density (i_{corr}) of the samples were obtained by fitting and analyzing the polarization curves (table plots) via Gamry Echem Analyst software. The inhibition efficiency (IE%) of the surface-functionalized samples was determined via Eq. (5):

$$\text{IE}\% = \frac{i_o - i_s}{i_o} \times 100\% \quad (5)$$

where i_o and i_s are the corrosion current densities of the original SS and modified samples, respectively, as determined by Tafel polarization curve analysis.

3 Results and discussion

3.1 Basic characterization

3.1.1 Morphological characterization

The electron micrographs of the hydrogel coating surfaces before water absorption are shown in Figs. 2(a)–2(c), and those after water absorption are shown in Figs. 2(d)–2(f). Combined with the three-dimensional contour plots in Figs. 2(a1)–2(f1), it is clear that all three hydrogel coating surfaces underwent hydration reactions after water absorption. The surface roughness (S_a) at the same locations before and after the formation of the hydrated layer was characterized via Keens Multi-File Analysis software, and the results showed that the S_a values of the PG/PVP, PG/PVP/PEI, and PG/PVP/PEI/V₂O₅ coatings were 23.168, 23.332, and 16.897 µm, respectively, indicating that the modification with VP

Table 2 Poisson's ratios and Young's moduli of diamond indenter and stainless steel substrate, and calculated values of reduced Young's modulus

Material	Poisson's ratio	Young's modulus (MPa)	Ref.
Diamond (ball indenter)	0.07	1,140	[32]
SS 304 (substrate)	0.3	210	[33]
Calculated reduced Young's modulus	—	192.0776	—

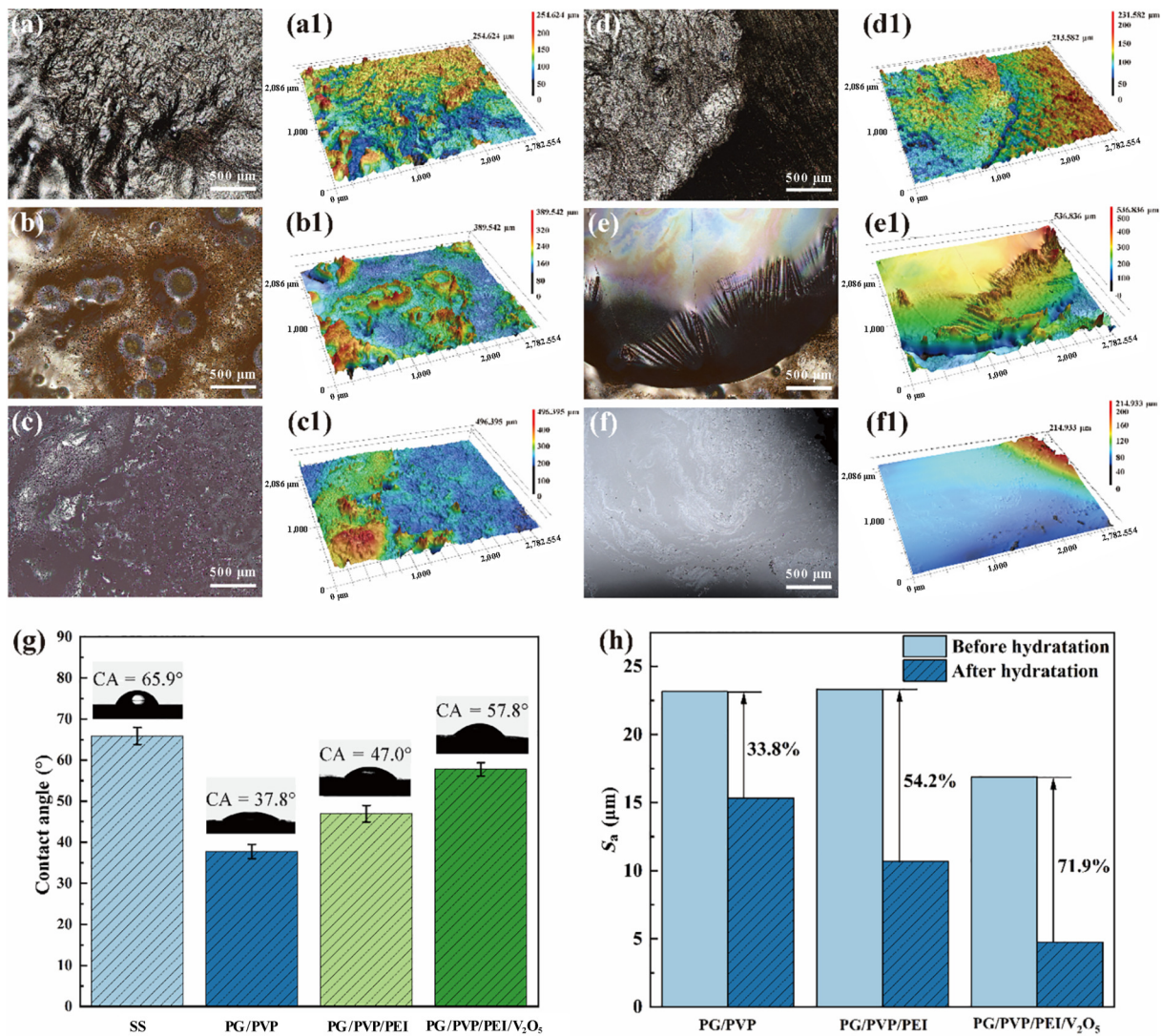


Fig. 2 Electron micrographs of dry PG/PVP (a), PG/PVP/PEI (b), and PG/PVP/PEI/V₂O₅ (c) surfaces; electron micrographs of wet PG/PVP (d), PG/PVP/PEI (e), and PG/PVP/PEI/V₂O₅ surfaces (f). Corresponding three-dimensional contour maps (a1–f1); contact angles of sample surfaces before and after crosslinking (g); and surface roughness (S_a) of samples before and after swelling (h).

made the surface of the coatings smoother (Fig. 2(h)). After full immersion, the S_a values of the three coatings decreased by 33.8%, 54.2%, and 71.9%, respectively, indicating that a smooth hydration layer was formed on the surface of the hydrogel coatings, which contributed to the antifouling, anticorrosive, and lubricating properties of the system.

The water contact angle of the sample surface corresponded to its wettability. A contact angle with water of 0° – 70° indicated that the coating was hydrophilic, and a smaller contact angle indicated a higher wettability of the coating [34, 35]. The surface water contact angle of the PG/PVP hydrogel coating was only $38^\circ \pm 0.4^\circ$, indicating that the coating was highly hydrophilic (Fig. 2(g)). The surface water contact angle of the PG/PVP/PEI hydrogel coating after PEI crosslinking increased to $47.5^\circ \pm 1.65^\circ$ because PEI is a polymer enriched with hydrophobic N-H groups. The VP-modified PG/PVP/PEI/V₂O₅ hydrogel coatings still exhibited hydrophilicity, and the surface hydrophobicity further increased to $58.9^\circ \pm 2.2^\circ$, which was attributed to the fact that coatings containing metal ions are usually hydrophobic.

SEM was used to characterize the surface microstructure of the freeze-dried hydrogel coating. The PG/PVP hydrogel coating was relatively smooth and flat (Figs. 3(a) and 3(a1)), whereas due to

the fact that the PG/PVP/PEI hydrogel had a richer lamellar structure, the uneven local water loss during dehydration led to the formation of a wavy surface structure (Figs. 3(b) and 3(b1)). The surface morphologies of the PG/PVP/PEI/V₂O₅ hydrogel in its original state after lyophilization and after 7 and 14 d of self-polishing showed the surface dissolved layer possessing spherical agglomerates after drying (Figs. 3(c) and 3(d)). The inside dried hydrated layer showed a relatively smooth surface, indicating that the water absorption and swelling of the PG/PVP/PEI/V₂O₅ hydrogel progressed from the outside to the inside. With the dissolution of the surface swelling layer, the inner smooth surface was exposed, which was a smooth surface with the same characteristics as those of self-polishing antifouling coatings. This phenomenon was based on the same principle as that of self-polishing marine antifouling coatings [29]. As the self-polishing process proceeded, the copolymers on the surface gradually dissolved and agglomerated, exposing the internal surface, and the above process was repeated over time.

3.1.2 FT-IR, XPS, and XRD characterization

FT-IR was used to collect information on the surface functional groups of the PVP-hydrated coating and the three hydrogel

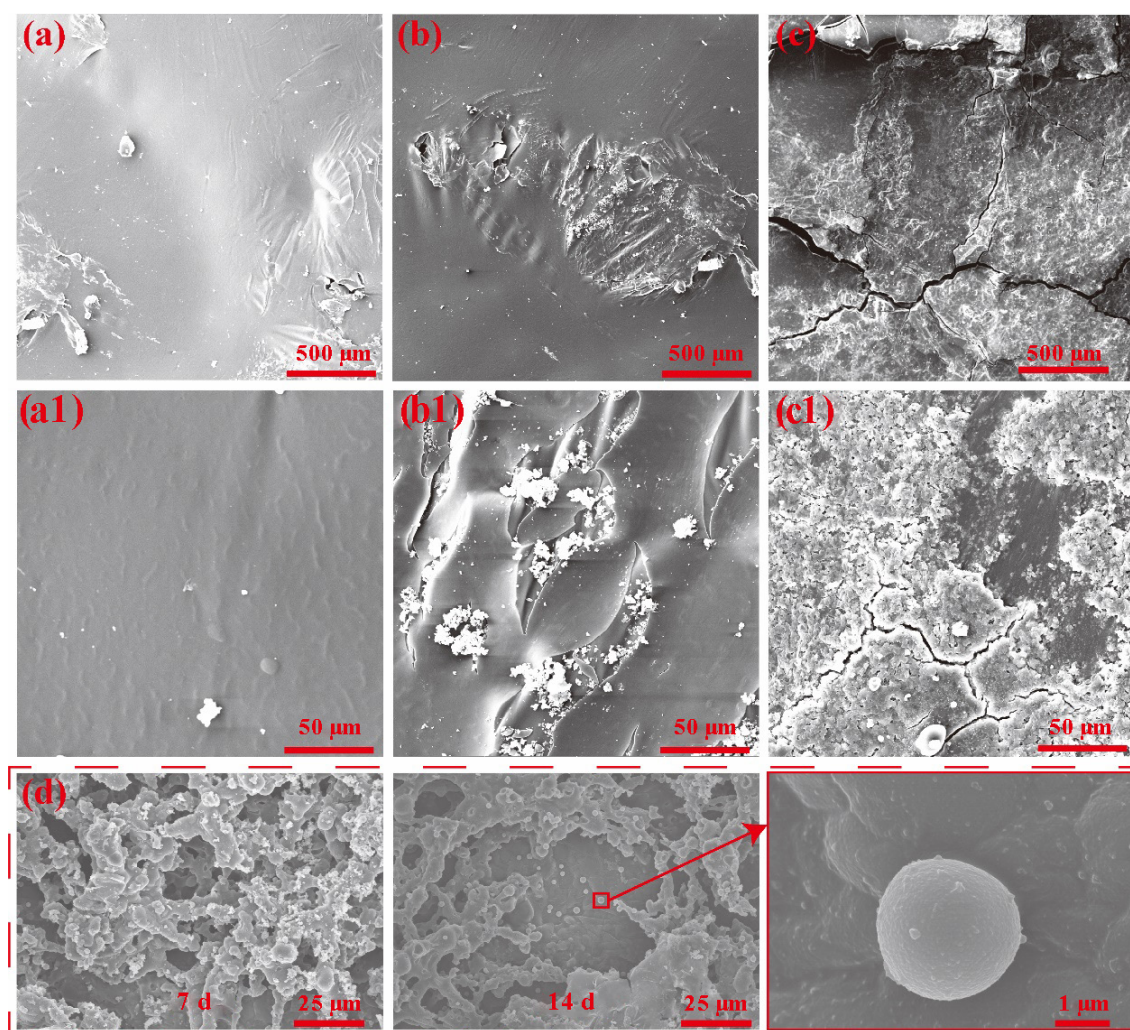


Fig. 3 SEM images of hydrogel-coated surfaces: PG/PVP (a), PG/PVP/PEI (b), and PG/PVP/PEI/V₂O₅ (c); typical surface morphologies of PG/PVP/PEI/V₂O₅ coatings after self-polishing (d).

coating samples. The absorption peaks of the Lewis basic groups C=O and C–N of the PVP-hydrated coating appeared at 1,639 and 1,290 cm^{−1}, respectively, whereas the absorption peak at 3,370 cm^{−1} was attributed to the hydroxyl peak produced by the hydrate (Fig. 4(a)). When PVP absorbed PG, the intensity of the C=O peak decreased significantly, and the C–N peak disappeared, indicating that the adsorption between PG and PVP was based on H-bonding, where PVP was the H-bond acceptor and PG was the H-bond donor. The FT-IR results of the three hydrogel coating systems revealed the presence of C=C stretching vibrations at 1,479 cm^{−1}, OH deformation vibrations at 1,290 cm^{−1}, and C–H stretching vibrations at 990 cm^{−1} induced by phenol, reflecting the molecular structure of the PG/PVP copolymer (Fig. 1). The absorption peak generated by N–H after the cross-linking of PEI appeared at 3,147 cm^{−1}. In addition, the C–O peak at 1,182 cm^{−1} indicated that a portion of the phenol was oxidized to benzoquinone, which was the reason for the stickiness of the hydrogel.

The sample surfaces were further analyzed by XPS (Fig. 4(b)), and the elemental composition obtained from the C 1s peak (284.8 eV) was corrected and fitted by Advantage software (Table 3). The results showed that PEI addition brought abundant N to the sample surface and promoted the Michael addition and Schiff base reactions between PEI and PG. In addition, VP was highly concentrated after drying of the hydrogel, with a high

content of 36.24 at%, which provided the necessary metal ions to further inhibit the proliferation of surface fouling organisms. Further fitting of the V 2p spectra revealed peaks induced by V⁵⁺ 2p_{3/2} and V⁵⁺ 2p_{1/2} at 517.2 and 524.5 eV, respectively, indicating that PG did not change the valence state of V after coordination (Fig. 4(c)) [36]. In addition, the characteristic V₂O₅ peaks at 2θ values of 20.1° and 42° in the XRD patterns of the PG/PVP/PEI/V₂O₅ coatings also indicated that V maintained its valence state during the reaction process (Fig. 4(d)) [37].

3.2 Swelling test

Swelling ratio measurements are widely used to characterize the water absorption and stability of hydrogel materials [38]. For hydrogels used in marine antifouling applications, high swelling ratios mean that they absorb more water and increase their weight, thus consuming more fuel. The swelling rate obtained from this swelling experiment is shown in Fig. 5(a). The results show that the uncrosslinked PG/PVP hydrogel reached swelling equilibrium only after 240 min, at which time the swelling rate was 13.56%, followed by significant dissolution, which was exhausted after 660 min. After the introduction of chemical crosslinking of PEI, the hydrogel coating reached swelling equilibrium after 360 min, and the swelling rate increased to 37.89%, indicating that the crosslinking resulted in a richer pore structure within the hydrogel coatings to accommodate more

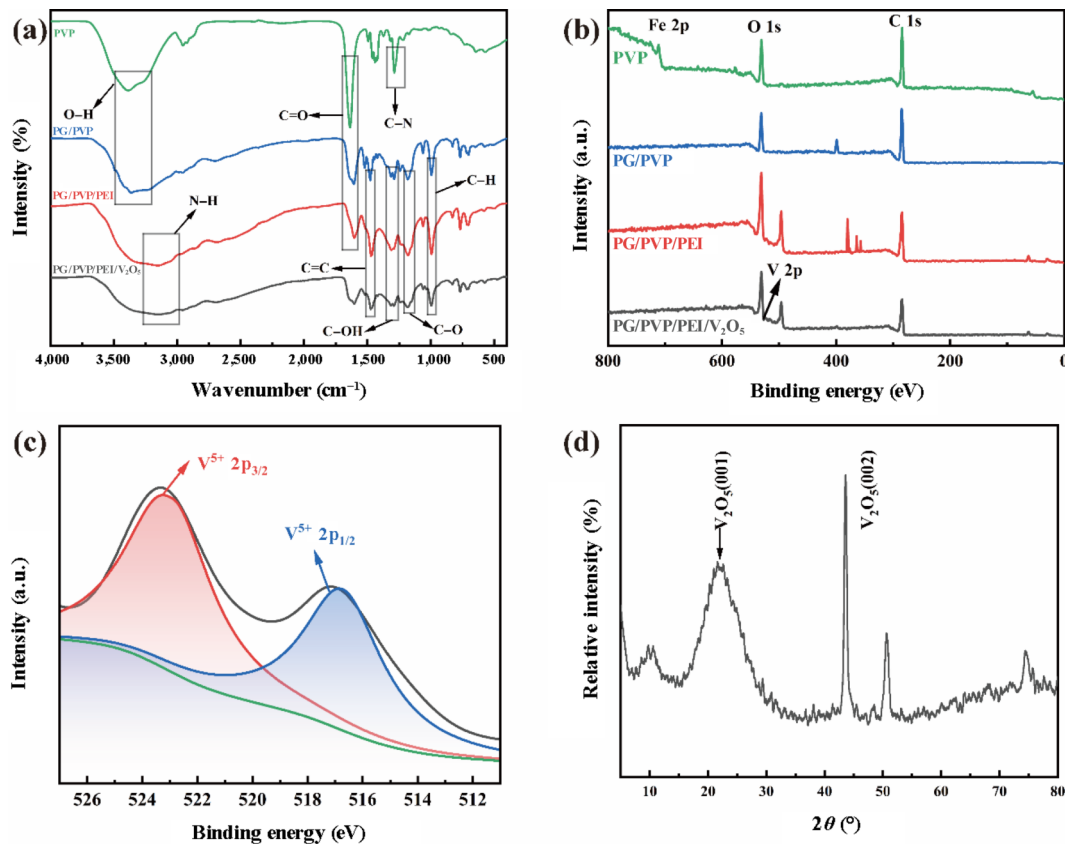


Fig. 4 Composition and valence characterization: FT-IR spectra (a), XPS survey spectra (b), V 2p spectra of PG/PVP/PEI/V₂O₅ (c), and XRD patterns of PG/PVP/PEI/V₂O₅ (d).

Table 3 XPS quantitative analysis results

Sample	C (at%)	N (at%)	O (at%)	V (at%)
PVP	48.25	5.46	46.30	N/A
PG/PVP	48.30	10.82	40.87	N/A
PG/PVP/PEI	36.80	12.63	50.56	N/A
PG/PVP/PEI/V ₂ O ₅	23.95	3.23	36.58	36.24

water. The swelling rate of the hydrogel coatings was reduced by VP modification, and the swelling rate was reduced to 32.36% after 420 min. This was attributed to the cross-linking of the metal sites of VP with the PG/PVP copolymer in the form of ligands, which resulted in a denser internal structure, with a higher cross-linking density resulting in a lower swelling rate. A comparison of the cross sections of the PG/PVP/PEI/V₂O₅ coating before swelling (Fig. 5(c)) and after swelling equilibrium (Fig. 5(d)) reveals that the coating thickness swells only from 53.51 to 72.90 μm without significant deformation, indicating that when the coating is applied to the surface of marine equipment, the marine equipment is less affected by the increase in drag force due to water absorption or paint deformation. The swelling curves demonstrate that the VP-modified hydrogels exhibit reduced solubility in artificial seawater due to the gradual hydrolysis of the primary and secondary chains of the hydrogel copolymers following prolonged immersion in solution. This process gives the coating surfaces the ability to continually flake off, thereby resulting in smoother surfaces. To determine the self-polishing rates of the PG/PVP/PEI and PG/PVP/PEI/V₂O₅ hydrogel coatings, 240-h dissolution experiments were conducted (Fig. 5(b)). The results show that the average dissolution rate of the

PG/PVP/PEI/V₂O₅ hydrogel in the static state every 12 h is only 0.011%, showing a long-lasting protection ability.

3.3 Lubricating properties

The COFs as a function of time for SS and hydrogels at different friction frequencies produced calculated averages (Table 4 and Figs. 6(a)–6(c)). At a friction frequency of 1 Hz, the PG/PVP hydrogel and the PG/PVP/PEI/V₂O₅ hydrogel had similar friction coefficients of 0.461 and 0.417, respectively, while the friction coefficient of the PG/PVP/PEI hydrogel increased with time and the final average friction coefficient was 0.650 (Fig. 1(a)). At 2 and 4 Hz, PG/PVP had favorable lubricity at the friction frequency. For the PG/PVP/PEI hydrogel, the overall COF tended to increase with time and fluctuated greatly in the friction experiments because it had the lowest modulus and crept upon contact, thus adhering to the specimen. PG/PVP/PEI/V₂O₅ had better lubrication properties than SS during seawater infiltration. The average COF values of PG/PVP/PEI/V₂O₅ were 0.417, 0.265, and 0.309 at 1, 2, and 4 Hz, respectively, and showed almost no fluctuation, which indicated more stable lubrication properties compared to 0.614, 0.601, and 0.585 for SS with seawater infiltration, respectively. Combined with the surface morphology

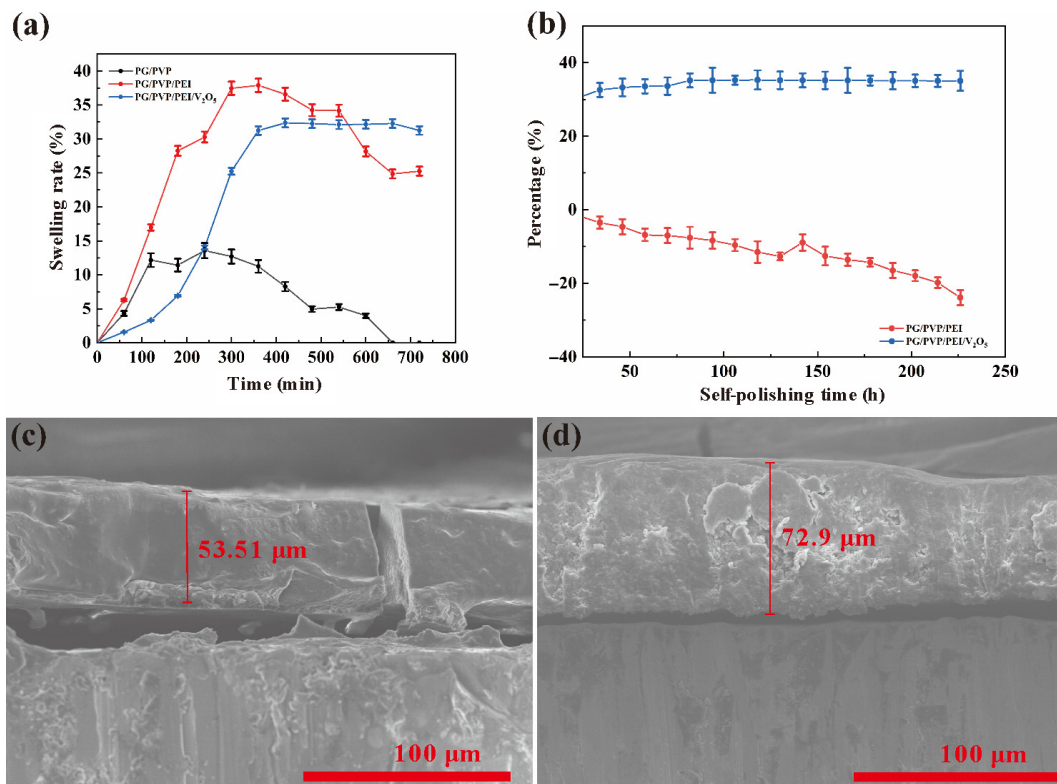


Fig. 5 Studies related to hydrogel swelling behavior: dissolution rate curve (a), self-polishing rate curve (b), comparison SEM images before (c) and after (d) swelling of PG/PVP/PEI/V₂O₅.

Table 4 Average COF of each surface

Sample	Frequency (Hz)	Average COF
SS	1	0.614
	2	0.601
	4	0.585
PG/PVP	1	0.461
	2	0.263
	4	0.294
PG/PVP/PEI	1	0.650
	2	0.966
	4	0.824
PG/PVP/PEI/V ₂ O ₅	1	0.417
	2	0.265
	4	0.309

of PG/PVP/PEI/V₂O₅ samples shown in SEM images above, this was attributed to the conversion of the sliding friction on the sample surfaces into rolling friction by the numerous spherical particles, which greatly improved the lubricating performance of the hydrogels (Fig. 6(d)).

3.4 Mechanical performance

3.4.1 Tensile test

The tensile method was used to investigate the mechanical properties of the PG/PVP hydrogels before and after chemical crosslinking with PEI and to study the effect of the addition of VP coordination crosslinking on the mechanical properties (Fig. 7(a)). For the PG/PVP hydrogel without the PEI crosslinking agent, the tensile modulus measured in the tensile test was 420 kPa, and the elongation at break was 1,622.4% (Fig. 7(b)). The tensile modulus

of the crosslinked PG/PVP/PEI hydrogel was only 208 kPa, but the elongation at break was up to 3,836.36%, which was almost double that before the addition of PEI (Fig. 7(c)). After the addition of the PEI crosslinking agent, more crosslinking points appeared in the hydrogel system, forming a more homogeneous network structure, which illustrated the excellent properties of PEI as a long-chain flexible crosslinker [39]. After modification by VP, the tensile modulus of the hydrogel increased to 744.07 kPa, and the elongation at break decreased to 894.48%, indicating that the chelating reaction between VP and the macromolecular network structure formed by the adsorption of PG with PVP increased the overall strength. Compared with the former, the coating made with the PG/PVP/PEI/V₂O₅ hydrogel had a smaller amount of creep under the drag of seawater.

3.4.2 Scratch testing results

The bond strength between the substrate and coating is a parameter that determines the durability and performance of the coating because of the positive correlation between the bond strength and force required to peel the coating [40]. To evaluate the bond strengths of the PG/PVP, PG/PVP/PEI, and PG/PVP/PEI/V₂O₅ coatings on the SS surfaces, a series of micron scratch tests were performed (Fig. 8). The critical load (L_c) refers to the normal force corresponding to the failure of the coating, and L_c can be further divided into the load at the time of cracking on both sides of the scratch (L_{c1}), at the time of peeling on both sides of the coating (L_{c2}) and at the time of complete peeling of the coating (L_{c3}) [41]. As the hydrogel coating was a highly plastic material, there was no peeling in the form of cracks during the scratching process. According to the methodology of Barnes et al. [42], the critical normal pressures at which the hydrogel coatings begin to separate as well as those at which they completely separate are denoted as L_{c1} and L_{c2} and are used as a basis for calculating the mean contact stress p_m at the time of coating

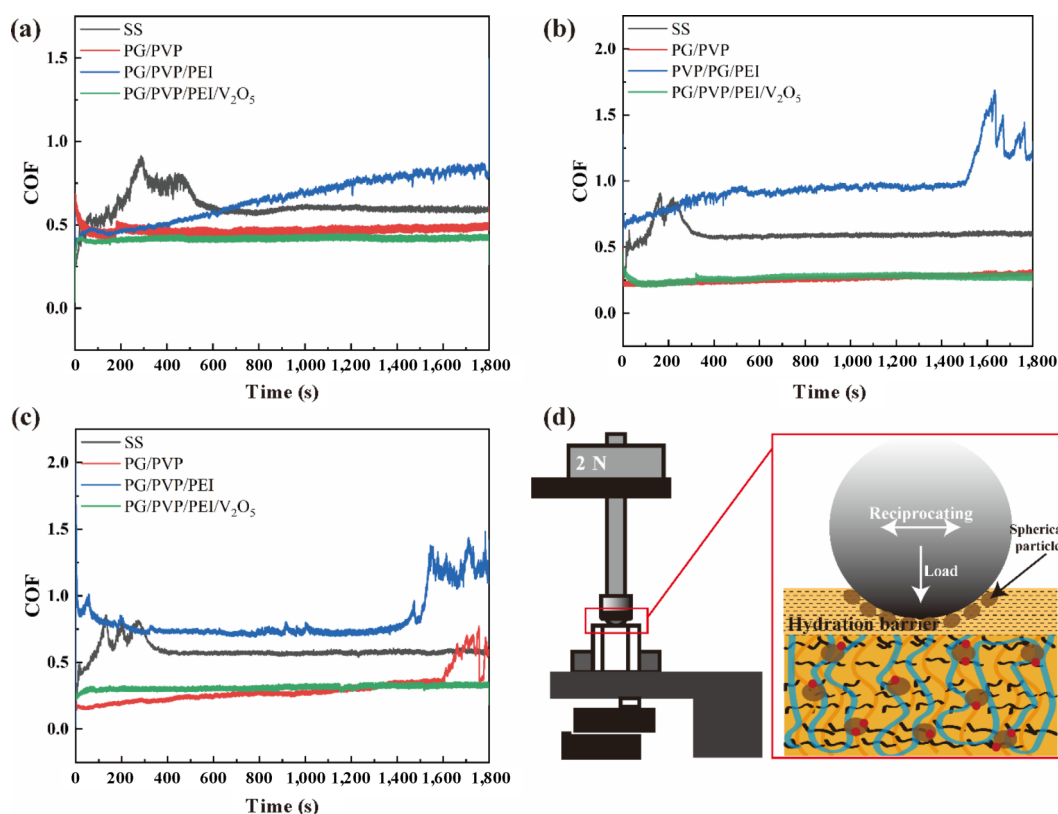


Fig. 6 Characterization of lubrication properties: COFs at 1, 2, and 4 Hz (a–c), respectively, and schematic diagram of friction experiment and lubrication mechanism (d).

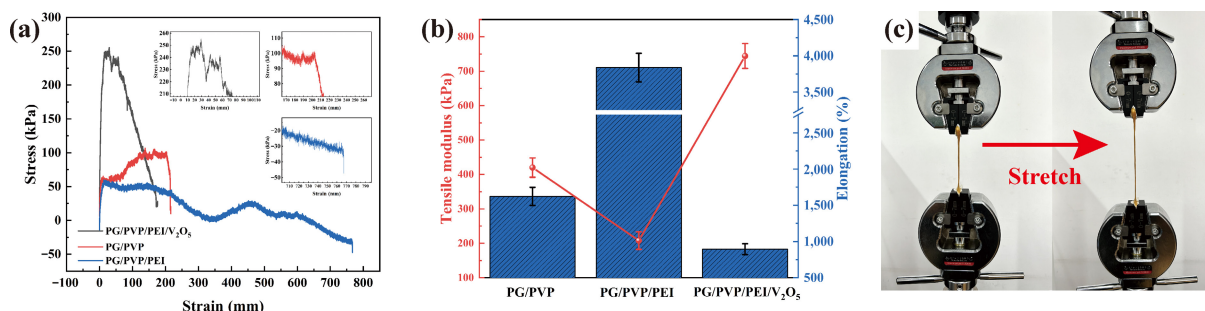


Fig. 7 Mechanical properties of three hydrogels: stress-strain curves (a), tensile moduli and elongations at break (b), and electronic photographs during tensile testing (c).

failure. During an experiment, an acoustic emission signal detector was used to receive the acoustic signals generated when the probe scraped the coating (Fig. 8(d)). When the acoustic signals changed abruptly, the probe caused the cohesion of the coating to fail, and the coating peeled off and rubbed directly against the SS 304 substrate.

The results of the scratching experiments on the three coatings revealed that they had similar scratching behaviors; when the indenter was in contact with the coating and did not scratch through the coating, the acoustic emission (AE) signals were almost unchanged due to the lower modulus of the hydrogel coating itself, while the AE signal began to gradually increase when the indenter started to scratch through the coating, and when the indenter completely stripped the coating, the intense friction with the substrate caused fluctuations in the AE signals (Figs. 8(a)–8(c)). The chemical crosslinking of PEI resulted in an increase in L_{c1} and L_{c2} from 3.93 and 4.06 to 4.13 and 4.19 N, respectively when the coating was fractured and further increased to 4.59 and 4.71 N after VP modification, indicating that both

types of crosslinking increased the mechanical stability of the coating in a stepwise manner. According to the critical mean pressure data obtained for L_{c1} and L_{c2} , as shown in Table 5, the adhesion of all three coatings reached more than 3 MPa, which means that their wet adhesion strength is greater than those of many reported gel-based adhesives [43–45].

3.5 Antifouling performance

Biofouling occurs in the early stages when microorganisms, such as bacteria, algae and biomolecules, colonize the surface of marine equipment [46]. As the PG/PVP coating lacks PEI as a crosslinking agent and self-polishes more rapidly in seawater, the test conditions were not available, and thus only the SS 304 substrate and the PG/PVP/PEI and PG/PVP/PEI/V₂O₅ hydrogel membranes were tested for fouling resistance. *S. aureus* and *V. natriegens* were chosen as representatives of marine gram-positive and gram-negative bacteria, respectively, for the antifouling experiments, while *C. pyrenoidosa* and *P. tricornutum* were chosen as representatives of marine Chlorophyta and

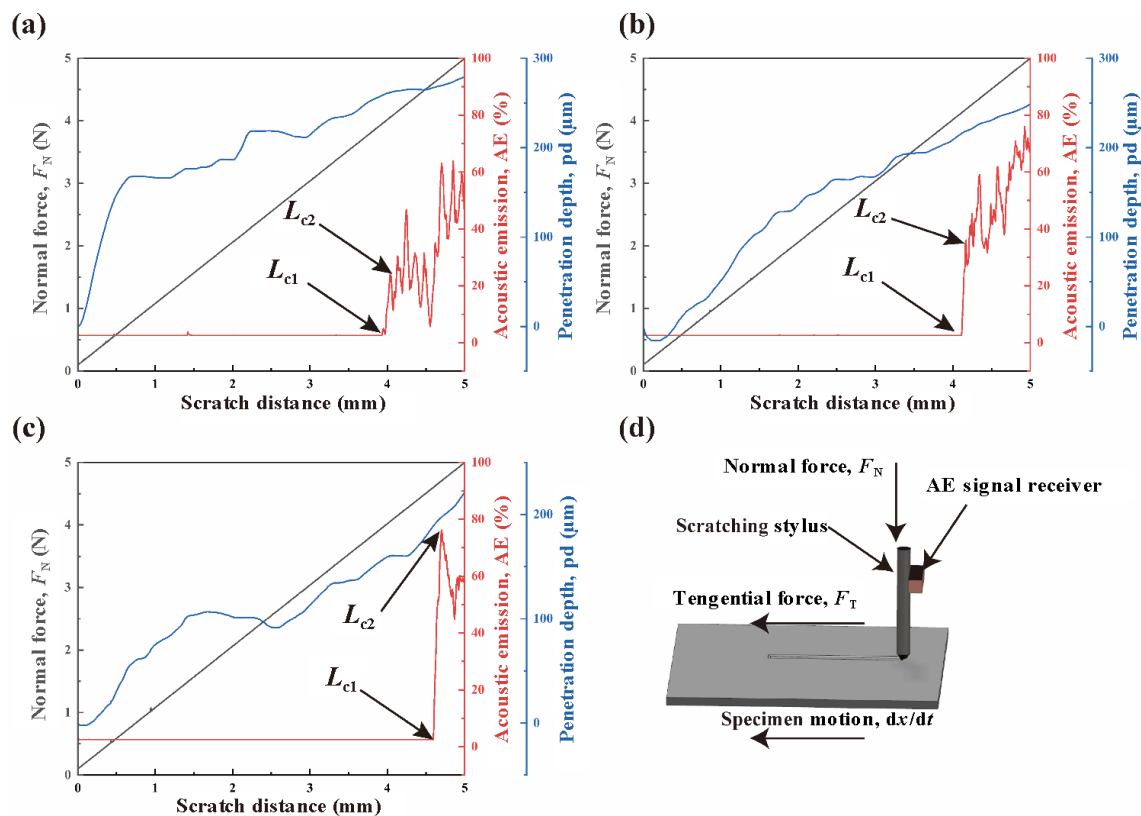


Fig. 8 Results of MST experiment for PG/PVP (a), PG/PVP/PEI (b), PG/PVP/PEI/V₂O₅ (c), and schematic diagram of MST principle (d).

Table 5 Values of L_{c1} and L_{c2} obtained from scratch test and critical mean pressure calculated using Eq. (3). $p_{m,c1}$ and $p_{m,c2}$ are the critical pressures for initial and total delamination, respectively

Sample	L_{c1} (N)	$p_{m,c1}$ (MPa)	L_{c2} (N)	$p_{m,c2}$ (MPa)
PG/PVP	3.93	3.21	4.06	3.25
PG/PVP/PEI	4.13	3.26	4.19	3.28
PG/PVP/PEI/V ₂ O ₅	4.59	3.37	4.71	3.42

Diatomophyta, *S. aureus*, *V. natriegens*, *C. pyrenoidosa*, and *P. tricornutum* are model bacterial and algal species that are widely used to study biofouling [47–50]. The rates of bacterial adhesion on sample surfaces are positively proportional to the rate at which biofouling occurs, and experiments were conducted using CLSM to determine the microbial adhesion rate on sample surfaces (Figs. 9(a) and 9(b)). The four fouling organisms adhered to the SS 304 surface at rates of 39.32%, 4.53%, 7.794%, and 4.63%, whereas for the PG/PVP/PEI surface, the inhibition rates of bacterial adhesion reached 99.96%, 87.75%, 86.60%, and 54.83%, respectively. In addition to the fact that the hydration layer on the hydrogel surface made it difficult for microbes to adhere, these results were attributed to the fact that PG, as a tannin, had an inhibitory effect on the growth of microorganisms, with some of them inhibited even at a concentration of 1 mM [51]. In PG/PVP/PEI hydrogels, PG is adsorbed by PVP to form polyphenols with stronger antimicrobial activity, while the quaternization of PEI inhibits bacterial growth through electrostatic interactions with bacterial cell membranes [52]. However, the results showed that the inhibitory effect of the PG/PVP/PEI hydrogels on diatoms was not strong. This occurred because the cell walls of diatoms contain silica, which acts as a shield for bacteriostatic substances. After the addition of VP to the PG/PVP/PEI hydrogel, the adhesion inhibition of the PG/PVP/PEI/V₂O₅ hydrogel film against the four microorganisms reached 99.97%, 99.77%, 99.87%, and 99.78%,

respectively (Fig. 9(c)). VP was observed to decrease the adhesion rate of the system to diatoms, which was due to the inhibitory effect of VP on biofilm formation, as do most metal ions [53]. In addition, the results of bacterial plate spreading on the surface before and after coating also showed that it was difficult for bacteria to grow on the surface of hydrogel coatings (Fig. 9(d)).

In conclusion, the designed PG/PVP/PEI/V₂O₅ hydrogel coatings strongly inhibited fouling organisms under the synergistic effects of PG, PEI, and vanadium pentoxide. Currently, antifouling coatings containing 5–20 wt% copper are widely used. However, it has been demonstrated that copper-containing antifouling agent particles (APPs) emanating from ship antifouling agents can cause reproductive toxicity to marine organisms as well as organisms in coastal soils [13, 54, 55]. Compared with current commercially available copper-based antifouling coatings, hydrogel coatings contain very low levels of metallic antifouling agents (≤ 1 wt%). This suggests that marine facilities coated with these coatings have a lower environmental load than traditional antifouling coatings do. Furthermore, PVP and PEI, which are common polymers in the chemical industry, do not accumulate in living organisms and cause ecological damage.

3.6 Corrosion inhibition analysis

Hydrophilic hydrogel coatings are considered protective, although

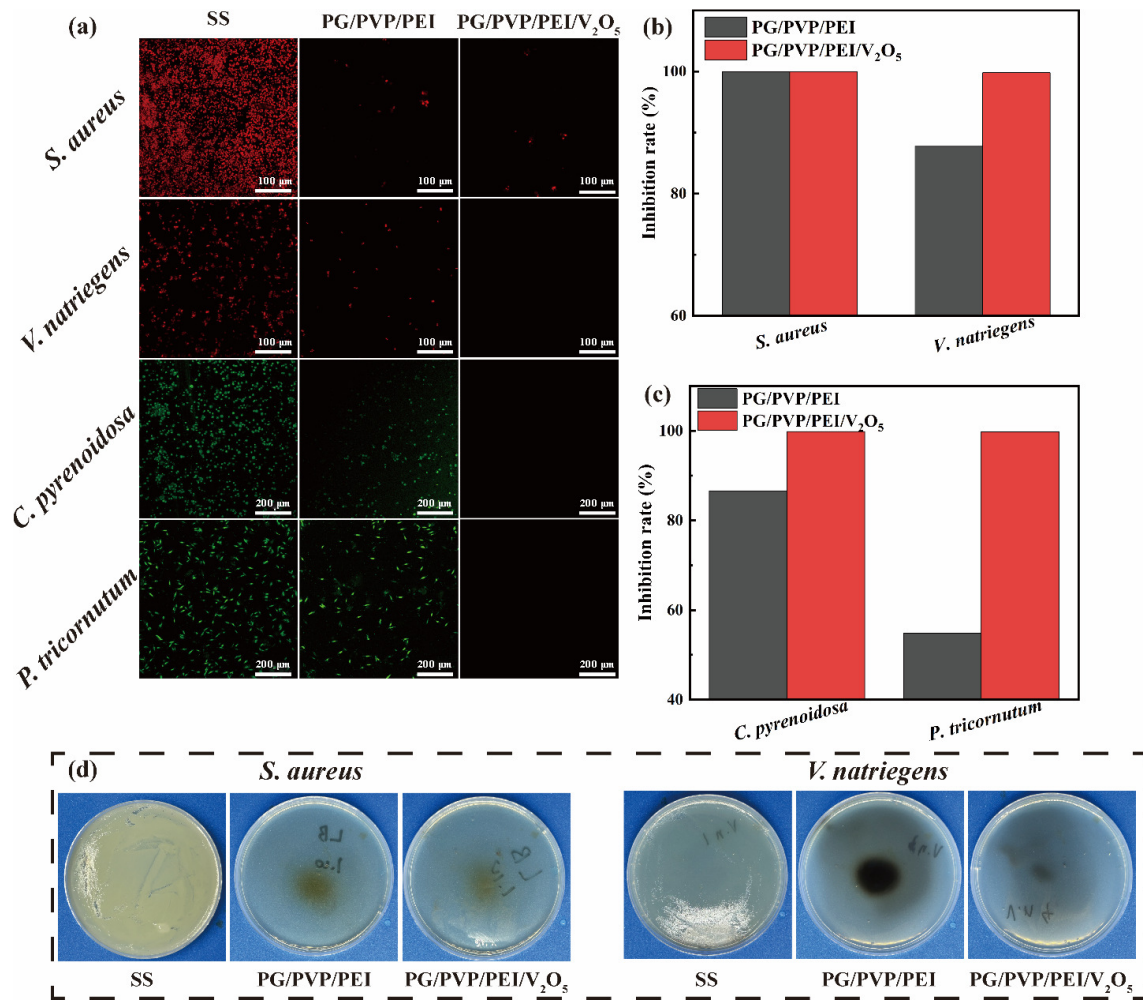


Fig. 9 Antifouling characterization: CLSM images of bacterial and algal adhesion on SS, PG/PVP/PEI, and PG/PVP/PEI/V₂O₅ surfaces (a), inhibition rates of bacterial adhesion on various surfaces (b), and inhibition rates of algal adhesion on various surfaces (c), *S. aureus* and *V. natriegens* test results of samples before and after coating for 5 h (d).

they do not completely isolate the substrate from direct contact with the electrolyte. The corrosion protection performance of the hydrogel-modified electrodes was analyzed by fitting the EIS and Tafel curves.

The corrosion resistance of the SS 304 substrate before and after hydrogel modification was analyzed by measuring the EIS of each electrode. For the Nyquist curves, the larger the radius of the impedance arc, the higher the charge transfer resistance on the metal surface, and the better the corrosion resistance [56]. The Nyquist curves for each electrode (Fig. 10(a)) had the SS 304 stainless steel electrode fitted using the equivalent circuit (Fig. 10(b1)) and the hydrogel-modified electrode fitted using the equivalent circuit (Fig. 10(b2)), where R_s , R_c , and R_{ct} represent the solution, coating, and charge transfer resistances, respectively. The constant-phase elements CPE_c and CPE_d represent the capacitances of the coating and bilayer, respectively. When the hydrogel coating swelled to mass balance, the coating was water-permeable but also considered to be protective, so only the constant phase element CPE_c was used for a more accurate fit [57]. PG/PVP and PG/PVP/PEI hydrogels had no positive effect on the corrosion mitigation of the SS 304 substrate, and the impedance arc radius of the PG/PVP/PEI hydrogel-modified electrode was significantly lower than that of the original samples because the PG/PVP/PEI hydrogel had the highest swelling rate. The impedance arc radius of the electrode, upon modification with the VP hydrogel, exhibited a notable increase compared with

that of the original sample, as deduced from the data presented in the preceding sections. This increase can be attributed to the augmented crosslinking strength of the hydrogel following the incorporation of VP, thereby intensifying the barrier effect against oxygen and ions.

For the Bode modulus and Bode phase angle plots, the impedance modulus in the low-frequency range ($|Z|_f = 0.01$ Hz) and the phase angle in the high-frequency range ($f = 10^4 - 10^5$ Hz) can reflect the corrosion resistance and shielding strength of the coating [58]. PG/PVP and PG/PVP/PEI were conductive and thus reduced the impedance modulus of the electrodes (Fig. 10(c)), while the VP-modified hydrogel electrode had the highest impedance modulus among the groups, which was $1.4144 \times 10^5 \Omega \cdot \text{cm}^2$ (9.8104×10^4 for unmodified SS 304). The phase angle diagrams indicated that SS 304 alone has good corrosion resistance, while the PG/PVP/PEI hydrogel exhibited fluctuations in the phase angle in the high-frequency region, which disappeared with the VP modification (Fig. 10(d)). Combining the above experimental results, PG/PVP/PEI had no corrosion protection effect on the SS surfaces while the PG/PVP/PEI/V₂O₅ hydrogel had better shielding performance and corrosion resistance.

The Tafel curve is a segment of the polarization curve in the highly polarized region, and the corrosion potential (E_{corr}) and corrosion current density (i_{corr}), obtained by fitting the polarization curve, can be used to characterize the rate at which corrosion is occurring. For E_{corr} and i_{corr} , the higher the value of

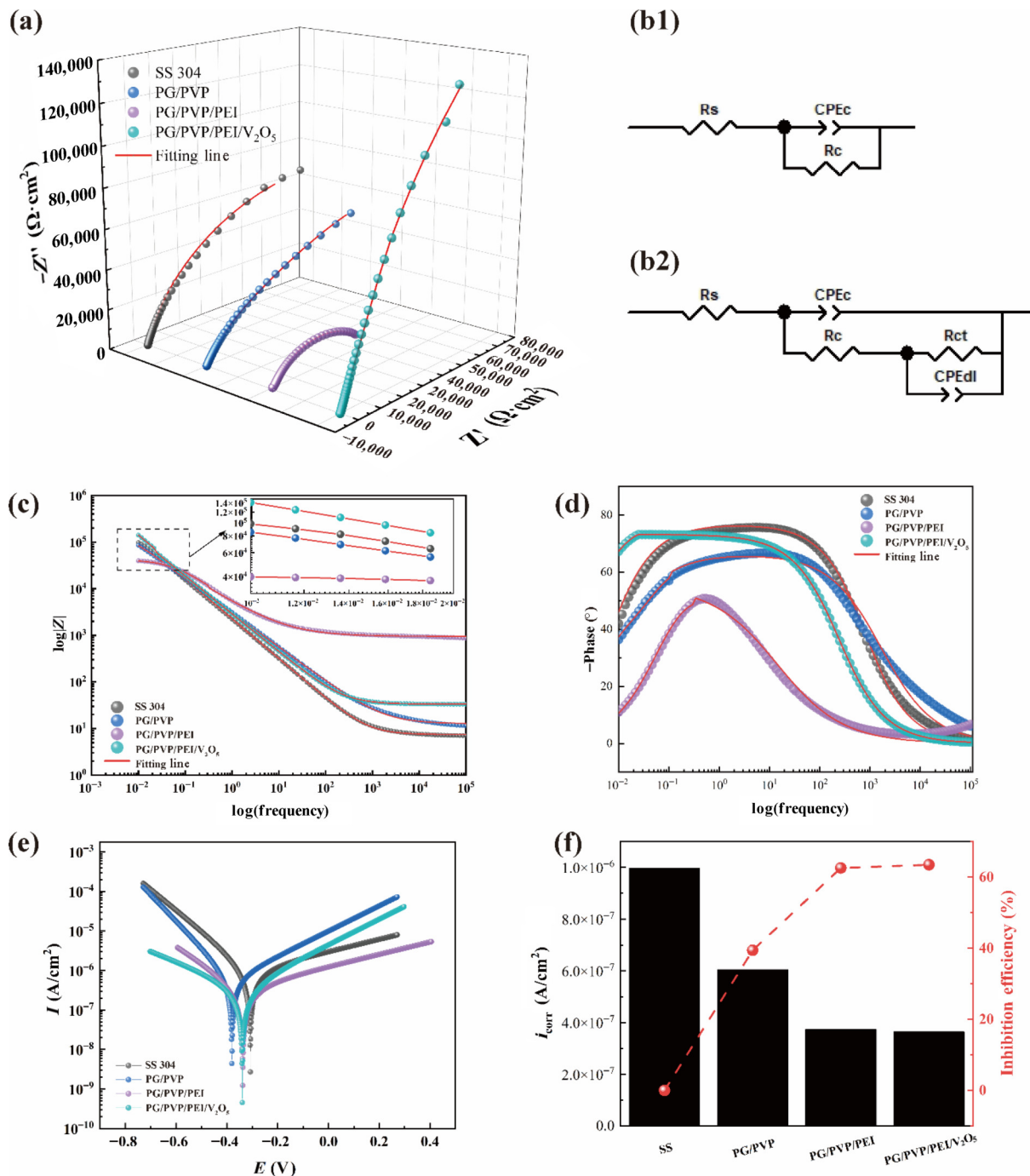


Fig. 10 Nyquist curves for all sample electrodes (a), equivalent circuit used to fit SS 304 electrodes (b1), equivalent circuit used to fit hydrogel electrodes (b2), Bode modulus plots (c), Bode phase diagrams (d), Tafel curves for all samples (e), and statistical plots of corrosion current densities and corrosion inhibition for all samples (f).

Table 6 Surface electrochemical corrosion parameters of each sample type

Sample	E_{corr} (V)	i_{corr} ($\mu\text{A}\cdot\text{cm}^{-2}$)	IE% (%)
SS	-0.308	0.997	—
SS-PG/PVP	-0.346	0.604	39.42
SS-PG/PVP/PEI	-0.445	0.445	62.59
SS-PG/PVP/PEI/V ₂ O ₅	-0.317	0.417	63.49

the former and the lower the value of the latter, the better the corrosion resistance of the sample. The Tafel curves for all the samples are shown in Fig. 10(e), and the data obtained via statistical fitting are presented in Table 6. Although the E_{corr} obtained by fitting increased for the PG/PVP and PG/PVP/PEI

hydrogel-modified electrodes, the surface corrosion currents of the samples gradually decreased with increasing PEI crosslinking (Fig. 10(f)). The corrosion inhibition of the coatings increased from 39.42% to 62.59%, whereas for the VP-modified hydrogel coatings, the value of E_{corr} increased from -0.445 to -0.317 mV, which was only 9 mV lower than that of SS 304, and the corrosion inhibition rate maintained at 63.49%. These findings indicated that the above hydrogel coatings on the SS 304 substrate surfaces reduced the corrosion current, while the corrosion potential of the metal surface covered with the hydrogel coatings was increased by the VP modification, which further improved the corrosion resistance of the samples.

The data in this section showed that the hydrogel coating

reduced the corrosion current density on the SS surfaces and thus retarded the corrosion rate due to the formation of a dissolved layer on the hydrogel surface after expansion, which acted as a barrier to oxygen and ions [59]. The PG/PVP/PEI/V₂O₅ hydrogel formed after VP modification showed the strongest corrosion inhibition performance, and, combined with the above water contact angle experimental results, this was due to the VP and hydrogel molecular branched chains of the benzene ring and hydroxyl coupling. This improved the densification and hydrophobicity of the coating to a certain extent, thus blocking the indirect contact between the NaCl solution and the substrate.

4 Conclusions

The purpose of this study was to develop a marine antifouling gelcoat that was easy to apply and can maintain adhesion in wet environments. A pyrogallol (PG)/polyvinylpyrrolidone (PVP) copolymer was prepared by PVP adsorption on phenols. A PG/PVP/polyethyleneimine (PEI)/V₂O₅ hydrogel was obtained by dual crosslinking of PG/PVP with PEI and V₂O₅. The performance of this gel and the coatings made from the gel were evaluated, and the experimental results are as follows:

(1) The PG/PVP copolymer acquired a water-absorbent swelling characteristic after chemical crosslinking with PEI, with a swelling rate (SR%) of 37% at swelling equilibrium, which was reduced to 32% upon co-crosslinking with VP.

(2) The PG/PVP/PEI hydrogels were co-crosslinked with VP to obtain stable lubricating properties, with the coefficient of friction (COF) reduced from 0.650, 0.966, and 0.824 to 0.461, 0.265, and 0.309 at 1, 2, and 4 Hz, respectively.

(3) After crosslinking with PEI, the p_{mc} to fully peel the coating was increased from 3.25 to 3.28 MPa, while further crosslinking of V⁵⁺ increased the p_{mc} to 3.42 MPa.

(4) PG/PVP/PEI/V₂O₅ coatings all achieved greater than 99% growth inhibition of typical fouling organisms, and the stainless steel (SS) corrosion inhibition increased to 63.49% after coating.

The experimental results revealed that the prepared hydrogel coatings had effective antifouling and anticorrosive properties and mechanical stability under humid conditions, which can open new research routes for the development of hydrogel-based coatings for marine antifouling.

Acknowledgements

This study was supported by the National Natural Science Foundation of China (Nos. 52275209 and 51905468), and the Jiangsu Province Graduate Student Practice and Innovation Program (No. SJCX23_1912).

Author contributions

Conceptualization: Pan Cao, Xiuqin Bai, and Chengqing Yuan; data curation: Pan Cao, Haobo Shu, and Junnan Cui; formal analysis: Haobo Shu; funding acquisition: Pan Cao, Xiuqin Bai, and Chengqing Yuan; investigation: Haobo Shu, Yuhan Liu, and Junnan Cui; experiment: Haobo Shu, Yuhan Liu, and Junnan Cui; software: Haobo Shu; supervision: Pan Cao and Chengqing Yuan. All the authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article. The author Chengqing Yuan is the Editorial Board Member of this journal.

References

[1] Banerjee I, Pangule R C, Kane R S. Antifouling coatings: Recent

developments in the design of surfaces that prevent fouling by proteins, bacteria, and marine organisms. *Adv Mater* 23(6): 690–718 (2011)

- [2] Jain A, Bhosle N B. Biochemical composition of the marine conditioning film: Implications for bacterial adhesion. *Biofouling* 25 (1): 13–19 (2009)
- [3] Aldred N, Clare A S. Mini-review: Impact and dynamics of surface fouling by solitary and compound ascidians. *Biofouling* 30(3): 259–270 (2014)
- [4] Fitridge I, Dempster T, Guenther J, de Nys R. The impact and control of biofouling in marine aquaculture: A review. *Biofouling* 28 (7): 649–669 (2012)
- [5] Farkas A, Degiuli N, Martić I. An investigation into the effect of hard fouling on the ship resistance using CFD. *Appl Ocean Res* 100: 102205 (2020)
- [6] Farkas A, Degiuli N, Martić I, Dejhalla R. Impact of hard fouling on the ship performance of different ship forms. *J Mar Sci Eng* 8(10): 748 (2020)
- [7] Anandkumar B, George R P, Maruthamuthu S, Parvathavarthini N, Mudali U K. Corrosion characteristics of sulfate-reducing bacteria (SRB) and the role of molecular biology in SRB studies: An overview. *Corros Rev* 34(1–2): 41–63 (2016)
- [8] Ashassi-Sorkhabi H, Moradi-Haghighi M, Zarrini G, Javaherdashti R. Corrosion behavior of carbon steel in the presence of two novel iron-oxidizing bacteria isolated from sewage treatment plants. *Biodegradation* 23(1): 69–79 (2012)
- [9] Beech I B, Sunner J. Biocorrosion: Towards understanding interactions between biofilms and metals. *Curr Opin Biotechnol* 15 (3): 181–186 (2004)
- [10] Negri A P, Heyward A J. Inhibition of coral fertilisation and larval metamorphosis by tributyltin and copper. *Mar Environ Res* 51(1): 17–27 (2001)
- [11] Villa L, D'Agati P, Mansueto C, Pellerito C, Scopelliti M, Fiore T, Nagy L, Pellerito L. Effects of tributyltin(IV) chloride on the gametes and fertilization of *Ascidia malaca* (Ascidacea: Tunicata). *Applied Organomet Chemis* 17(2): 106–112 (2003)
- [12] Nehring S. Long-term changes in Prosobranchia (Gastropoda) abundances on the German North Sea coast: The role of the anti-fouling biocide tributyltin. *J Sea Res* 43(2): 151–165 (2000)
- [13] Tian R M, Wang Y, Bougouffa S, Gao Z M, Cai L, Zhang W P, Bajic V, Qian P Y. Effect of copper treatment on the composition and function of the bacterial community in the sponge *Haliclona cymaeformis*. *mBio* 5(6): e01980–14 (2014)
- [14] Katranitsas A, Castritsi-Catharios J, Persoone G. The effects of a copper-based antifouling paint on mortality and enzymatic activity of a non-target marine organism. *Mar Pollut Bull* 46(11): 1491–1494 (2003)
- [15] He X Y, Cao P, Tian F, Bai X Q, Yuan C Q. Infused configurations induced by structures influence stability and antifouling performance of biomimetic lubricant-infused surfaces. *Surf Coat Tech* 358: 159–166 (2019)
- [16] Nurioglu A G, de With G. Non-toxic, non-biocide-release antifouling coatings based on molecular structure design for marine applications. *J Mater Chem B* 3(32): 6547–6570 (2015)
- [17] Waite J H. Mussel adhesion—essential footwork. *J Exp Biol* 220(4): 517–530 (2017)
- [18] Silverman H G, Roberto F F. Understanding marine mussel adhesion. *Mar Biotechnol* 9(6): 661–681 (2007)
- [19] Waite J H, Tanzer M L. Polyphenolic substance of *Mytilus edulis*: Novel adhesive containing L-dopa and hydroxyproline. *Science* 212 (4498): 1038–1040 (1981)
- [20] Yang Z L, Yang Y, Zhang L, Xiong K Q, Li X Y, Zhang F, Wang J, Zhao X, Huang N. Mussel-inspired catalytic selenocystamine-dopamine coatings for long-term generation of therapeutic gas on cardiovascular stents. *Biomaterials* 178: 1–10 (2018)
- [21] North M A, Del Grosso C A, Wilker J J. High strength underwater



- bonding with polymer mimics of mussel adhesive proteins. *ACS Appl Mater Inter* 9(8): 7866–7872 (2017)
- [22] Wang Y, Jeon E J, Lee J, Hwang H, Cho S W, Lee H. A phenol-amine superglue inspired by insect sclerotization process. *Adv Mater* 32(43): 2002118 (2020)
- [23] Jones T A, Wilker J J. Influences of phosphates on the adhesion of a catechol-containing polymer. *ACS Appl Polym Mater* 2(11): 4632–4639 (2020)
- [24] Gan D L, Xing W S, Jiang L L, Fang J, Zhao C C, Ren F Z, Fang L M, Wang K F, Lu X. Plant-inspired adhesive and tough hydrogel based on Ag-Lignin nanoparticles-triggered dynamic redox catechol chemistry. *Nat Commun* 10(1): 1487 (2019)
- [25] Shui T, Pan M F, Li A, Fan H B, Wu J P, Liu Q, Zeng H B. Poly(vinyl alcohol) (PVA)-based hydrogel scaffold with isotropic ultratoughness enabled by dynamic amine–catechol interactions. *Chem Mater* 34(19): 8613–8628 (2022)
- [26] Park H, Kim D. A rapid hydrophilization of porous poly-(tetrafluoroethylene) film via co-deposition of phenol derivatives and polyethyleneimine. *Prog Org Coat* 157: 106077 (2021)
- [27] Folch-Cano C, Olea-Azar C, Speisky H. Structural and thermodynamic factors on the adsorption process of phenolic compounds onto polyvinylpyrrolidone. *Colloid Surface A* 418: 105–111 (2013)
- [28] Li X R, Li J, Wang J Y, Yuan J, Jiang F, Yu X Y, Xiao F P. Recent applications and developments of Polyurethane materials in pavement engineering. *Constr Build Mater* 304: 124639 (2021)
- [29] Sha J N, Yu J, Chen R R, Liu Q, Liu J Y, Zhu J H, Liu P L, Li R M, Wang J. Eco-friendly self-polishing antifouling coating via eugenol ester hydrolysis. *Prog Org Coat* 172: 107077 (2022)
- [30] Johnson K L. *Contact Mechanics*. Cambridge (UK): Cambridge University Press, 1987.
- [31] Oliver W C, Pharr G M. An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments. *J Mater Res* 7(6): 1564–1583 (1992)
- [32] Zhang J J, Sun T, Hartmaier A, Yan Y D. Atomistic simulation of the influence of nanomachining-induced deformation on subsequent nanoindentation. *Comput Mater Sci* 59: 14–21 (2012)
- [33] Elkins J, Marsh J L, Lujan T, Peindl R, Kellam J, Anderson D D, Lack W. Motion predicts clinical callus formation: Construct-specific finite element analysis of supracondylar femoral fractures. *J Bone Joint Surg Am* 98(4): 276–284 (2016)
- [34] Katona G, Sipos B, Ambrus R, Csóka I, Szabó-Révész P. Characterizing the drug-release enhancement effect of surfactants on megestrol-acetate-loaded granules. *Pharmaceuticals* 15(2): 113 (2022)
- [35] Muhammed N S, Haq B, Al Shehri D, Al-Ahmed A, Rahman M M, Zaman E. A review on underground hydrogen storage: Insight into geological sites, influencing factors and future outlook. *Energy Rep* 8: 461–499 (2022)
- [36] Silversmit G, Depla D, Poelman H, Marin G B, De Gryse R. Determination of the V 2p XPS binding energies for different vanadium oxidation states (V^{5+} to V^{0+}). *J Electron Spectrosc* 135(2–3): 167–175 (2004)
- [37] Wang X J, Li H D, Fei Y J, Wang X, Xiong Y Y, Nie Y X, Feng K A. XRD and Raman study of vanadium oxide thin films deposited on fused silica substrates by RF magnetron sputtering. *Appl Surf Sci* 177 (1–2): 8–14 (2001)
- [38] Peng H T, Martineau L, Shek P N. Hydrogel–elastomer composite biomaterials: 1. Preparation of interpenetrating polymer networks and *in vitro* characterization of swelling stability and mechanical properties. *J Mater Sci—Mater M* 18(6): 975–986 (2007)
- [39] Shen J H, Zhang H, Zhu J X, Ma Y L, He H W, Zhu F B, Jia L, Zheng Q. Simple preparation of a waterborne polyurethane crosslinked hydrogel adhesive with satisfactory mechanical properties and adhesion properties. *Front Chem* 10: 855352 (2022)
- [40] Singla A, Singh N K, Singh Y, Jangir D K. Micro and nano-crystalline diamond coatings of co-cemented tungsten carbide tools with their characterization. *J Bio Tribo Corros* 7(2): 35 (2021)
- [41] Kim H K, Kim S M, Lee S Y. Mechanical properties and thermal stability of CrZrN/CrZrSiN multilayer coatings with different bilayer periods. *Coatings* 12(7): 1025 (2022)
- [42] Barnes D, Johnson S, Snell R, Best S. Using scratch testing to measure the adhesion strength of calcium phosphate coatings applied to poly(carbonate urethane) substrates. *J Mech Behav Biomed Mater* 6: 128–138 (2012)
- [43] Liang M, He C P, Dai J D, Ren P F, Fu Y F, Wang F M, Ge X, Zhang T Z, Lu Z H. A high-strength double network polydopamine nanocomposite hydrogel for adhesion under seawater. *J Mater Chem B* 8(36): 8232–8241 (2020)
- [44] Fu Y F, Ren P F, Wang F M, Liang M, Hu W J, Zhou N Z, Lu Z H, Zhang T Z. Mussel-inspired hybrid network hydrogel for continuous adhesion in water. *J Mater Chem B* 8(10): 2148–2154 (2020)
- [45] Liu Y Q, Wang P D, Su X, Xu L, Tian Z L, Wang H, Ji G J, Huang J Y. Electrically programmable interfacial adhesion for ultrastrong hydrogel bonding. *Adv Mater* 34(13): 2108820 (2022)
- [46] Yoon H, Baek Y, Yu J, Yoon J. Biofouling occurrence process and its control in the forward osmosis. *Desalination* 325: 30–36 (2013)
- [47] Lee H J, Kim H E, Lee C H. Combination of cupric ion with hydroxylamine and hydrogen peroxide for the control of bacterial biofilms on RO membranes. *Water Res* 110: 83–90 (2017)
- [48] Liu Q, Qiu G L, Zhou Z Z, Li J G, Amy G L, Xie J P, Lee J Y. An effective design of electrically conducting thin-film composite (TFC) membranes for bio and organic fouling control in forward osmosis (FO). *Environ Sci Technol* 50(19): 10596–10605 (2016)
- [49] Yoshida K, Tashiro Y, May T, Okabe S. Impacts of hydrophilic colanic acid on bacterial attachment to microfiltration membranes and subsequent membrane biofouling. *Water Res* 76: 33–42 (2015)
- [50] Liu H W, Sharma M, Wang J L, Cheng Y F, Liu H F. Microbiologically influenced corrosion of 316L stainless steel in the presence of *Chlorella vulgaris*. *Int Biodeter Biodegr* 129: 209–216 (2018)
- [51] Kocaalişkan I, Talan I, Terzi I. Antimicrobial activity of catechol and pyrogallol as allelochemicals. *Z Naturforsch C* 61(9–10): 639–642 (2006)
- [52] Wirde M, Gelius U, Nyholm L. Self-assembled monolayers of cystamine and cysteamine on gold studied by XPS and voltammetry. *Langmuir* 15(19): 6370–6378 (1999)
- [53] Wahid F, Zhong C, Wang H S, Hu X H, Chu L Q. Recent advances in antimicrobial hydrogels containing metal ions and metals/metal oxide nanoparticles. *Polymers* 9(12): 636 (2017)
- [54] Srinivasan M, Swain G W. Managing the use of copper-based antifouling paints. *Environ Manage* 39(3): 423–441 (2007)
- [55] Soroldoni S, Honscha L C, Reis F O, Duarte F A, da Silva Jr F M R, Pinho G L L. Antifouling paint particles in soils: Toxic impact that goes beyond the aquatic environment. *Ecotoxicology* 30(6): 1161–1169 (2021)
- [56] Cao C N. On the impedance plane displays for irreversible electrode reactions based on the stability conditions of the steady-state—I. One state variable besides electrode potential. *Electrochim Acta* 35(5): 831–836 (1990)
- [57] Wen J X, Lei J L, Chen J L, Gou J J, Li Y, Li L J. An intelligent coating based on pH-sensitive hybrid hydrogel for corrosion protection of mild steel. *Chem Eng J* 392: 123742 (2020)
- [58] Wei H G, Ding D W, Wei S Y, Guo Z H. Anticorrosive conductive polyurethane multiwalled carbon nanotube nanocomposites. *J Mater Chem A* 1(36): 10805–10813 (2013)
- [59] Yang H, Qin L G, Zhao W H, Mawignon F J, Guo H, Wu Y K, Zhang Y L, Dong G N. Eco-friendly polysaccharide coatings for antifouling and drag-reduction and potential application for marine devices. *Friction* 12(4): 726–744 (2024)



Haobo Shu received his B.S. degree in mechanical engineering from Yancheng Institute of Technology, Yancheng, China, in 2018. After that, he became a master candidate at the Institute of Surface and Interface Engineering, School of Mechanical Engineering, Yangzhou University, Yangzhou, China. His research is on environmentally friendly marine antifouling coatings.



Pan Cao received his B.S. degree in marine engineering in 2012 from Zhejiang Ocean University, Zhoushan, China. He received his M.S. degree in ship and ocean engineering and Ph.D. degree in transportation engineering from Wuhan University of Technology, Wuhan, China, in 2018. His research interest focuses on ship antifouling and drag reduction technology and green ship technology.



Chengqing Yuan received his B.S. degree in chemical engineering in 1998 from Wuhan University of Automobile Technology, Wuhan, China. He received his M.S. degree in mechanical design and theory at Wuhan Research Institute of Materials Protection, China Academy of Machinery Science and Technology, Wuhan, China, in 2001 and Ph.D. degree in vehicle utilization engineering from Wuhan University of Technology, Wuhan, China, in 2005. His current position is a professor and vice dean at the School of Transportation and Logistics Engineering, Wuhan University of Technology, in 2005. His research areas cover marine tribology, new energy aided ship power system, and ship energy efficiency improvement.



Xiuqin Bai is currently a professor at the School of Transportation and Logistics Engineering, Wuhan University of Technology, Wuhan, China. She obtained her M.S. degree at Wuhan Transportation University, Wuhan, China, in 1997. She received her Ph.D. degree from Wuhan Material Protection Research Institute, Wuhan, China, in 2006. Her research interest focuses on ship antifouling and drag reduction technology and green ship technology.